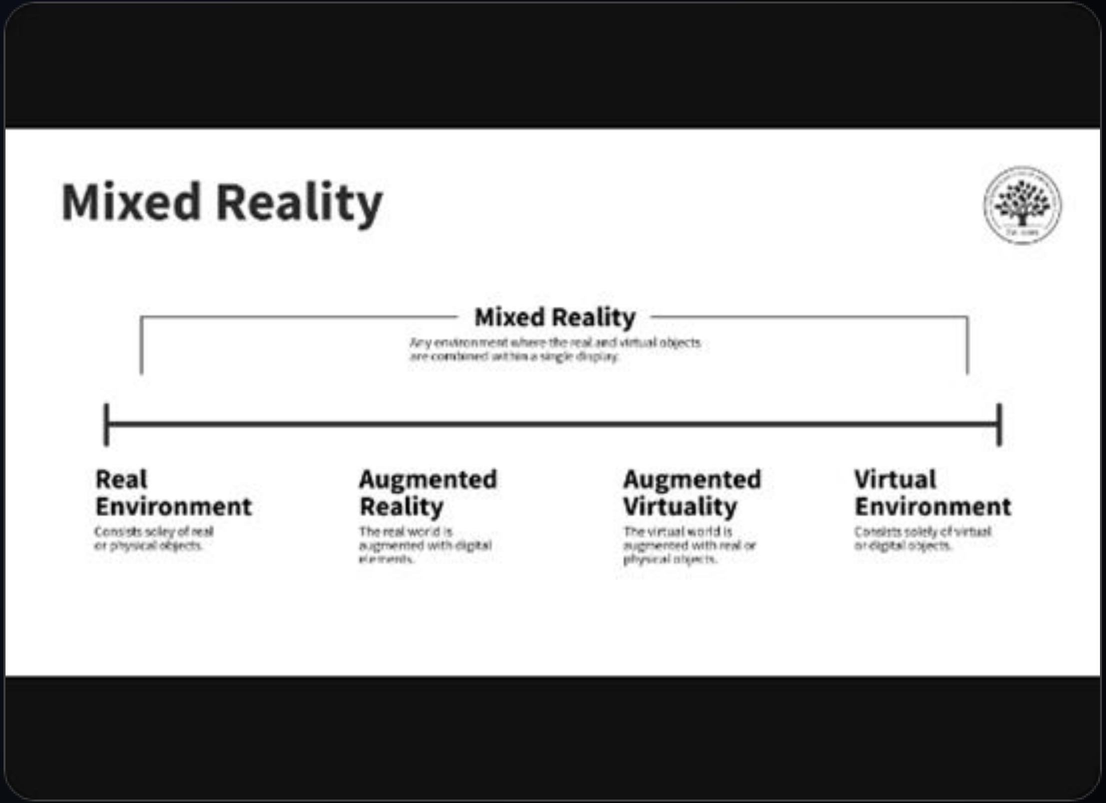
Augmented Reality

Milgram's Reality-Virtuality Continuum

The Spectrum of Experience

Paul Milgram's continuum (1994) defines the landscape of mixed reality. It is not a binary choice between Real and Virtual, but a continuous scale.

- **Real Environment:** Unmediated physical world.
- **Augmented Reality (AR):** Virtual elements overlaid on the real world.
- **Augmented Virtuality (AV):** Real elements overlaid on a virtual world.
- **Virtual Reality (VR):** Fully synthetic environment.



Taxonomy: AR, MR, and XR

Augmented Reality (AR)

Enhances the real world with digital data. The user maintains a sense of presence in the physical world. (e.g., Pokémon GO, Google Glass).

Mixed Reality (MR)

Virtual and real objects coexist and interact in real-time. Often implies a higher degree of environmental understanding. (e.g., HoloLens).

Extended Reality (XR)

The umbrella term encompassing AR, VR, MR, and all future immersive technologies.

Core Engineering Challenges

Registration

The accurate alignment of the virtual coordinate system with the real-world coordinate system. This is the "hard problem" of AR. Without precise registration, the illusion breaks.

Interaction

How the user inputs commands and manipulates virtual content. Unlike 2D screens, AR requires 3D spatial interaction, often using hands, gaze, or voice.

PART I

Hardware Architecture

The AR Pipeline

Sensing

Cameras, IMUs, GPS, Depth Sensors capture the environment.

Tracking

Algorithms (SLAM, VO) estimate pose (position & orientation).

Rendering

Graphics engine generates the virtual image based on pose.

Display

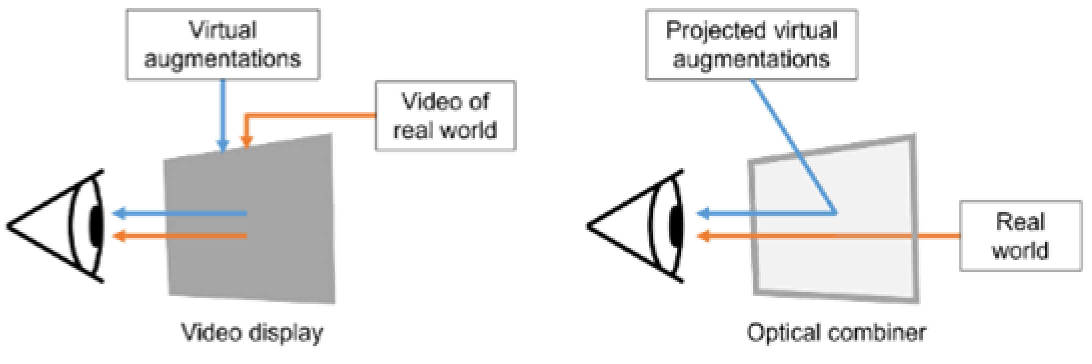
Optical or Video combination of real and virtual.

Display Technologies

See-Through Paradigms

AR displays are fundamentally categorized by how they combine the real and virtual light paths.

- **Optical See-Through (OST):** User sees the real world directly through optical combiners (mirrors, waveguides).
- **Video See-Through (VST):** Cameras capture the world, processed video is shown on opaque screens (e.g., Apple Vision Pro).



(a) Video see-through display

(b) Optical see-through display

Optical See-Through (OST)

- **Mechanism:** Uses partially transmissive mirrors or diffractive waveguides to overlay light.
- **Pros:** Zero latency for the real world (safety), natural resolution of the eye, lighter weight / form factor.
- **Cons:** Cannot display opaque black (black is transparent), complex calibration, limited field of view (FOV).
- **Key Tech:** Diffractive Waveguides, Birdbath Optics, Holographic Combiners.

Video See-Through (VST)

- **Mechanism:** Cameras capture reality; software composites virtual objects; displayed on opaque VR screens.
- **Pros:** Perfect occlusion (virtual can hide real), easier calibration, wide FOV, absolute control over every pixel.
- **Cons:** Photon-to-photon latency (safety risk), lower resolution than human eye, potential motion sickness.
- **Key Tech:** Low-latency passthrough cameras, high-res micro-OLEDs.

Sensing Modalities

Head Tracking

Gaze Tracking

Physiological Signals

creates a digital map of the physical space for realistic and oction.

Body Tracking

Gesture Recognition

Used for manalation and gesture function and recognition.

Environment Tracking

Creates a digital map of the physical space for realistic placement and occlusion.

Sensor Suite

IMU

Inertial Measurement Unit. High update rate (1000Hz+). Measures angular velocity (Gyroscopes) and linear acceleration (Accelerometers).

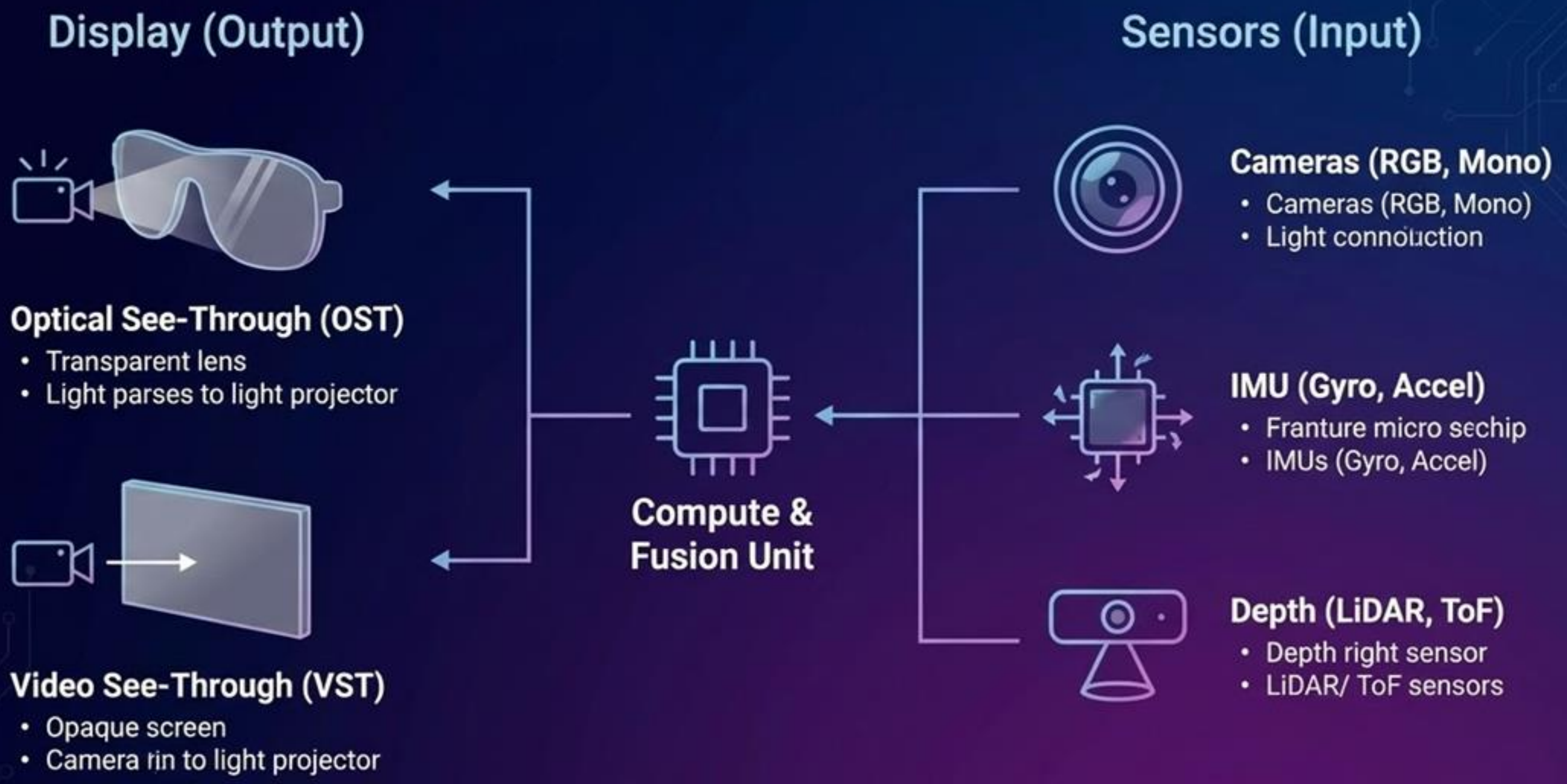
Vision

RGB and Monochrome cameras. Used for feature tracking, SLAM, and environmental understanding.

Depth

LiDAR (Light Detection and Ranging) or Time-of-Flight (ToF). Provides direct dense depth maps for occlusion and meshing.

AR Display & Sensor Integration



PART II

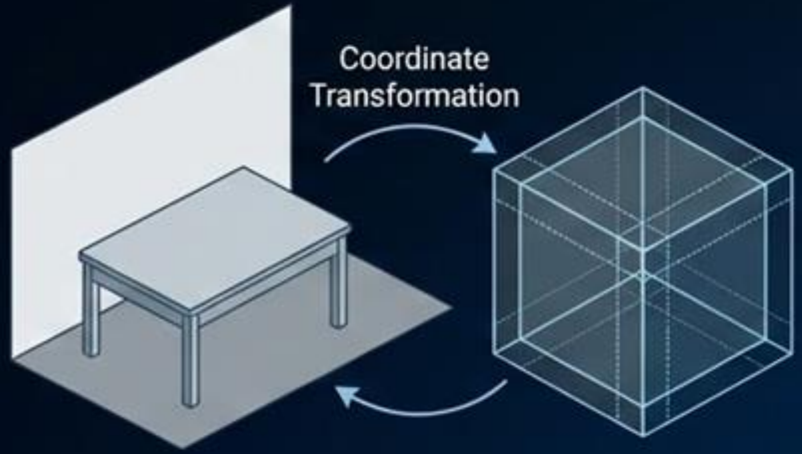
Registration & Tracking

AR Registration: The Foundation

Aligning the Virtual and Real Worlds

What is Registration?

- The process of accurately aligning virtual objects with the real-world coordinate system.
- Enables virtual content to appear fixed in physical space.
- Crucial for maintaining the illusion of a unified reality.



The "Hard Problem" Challenges

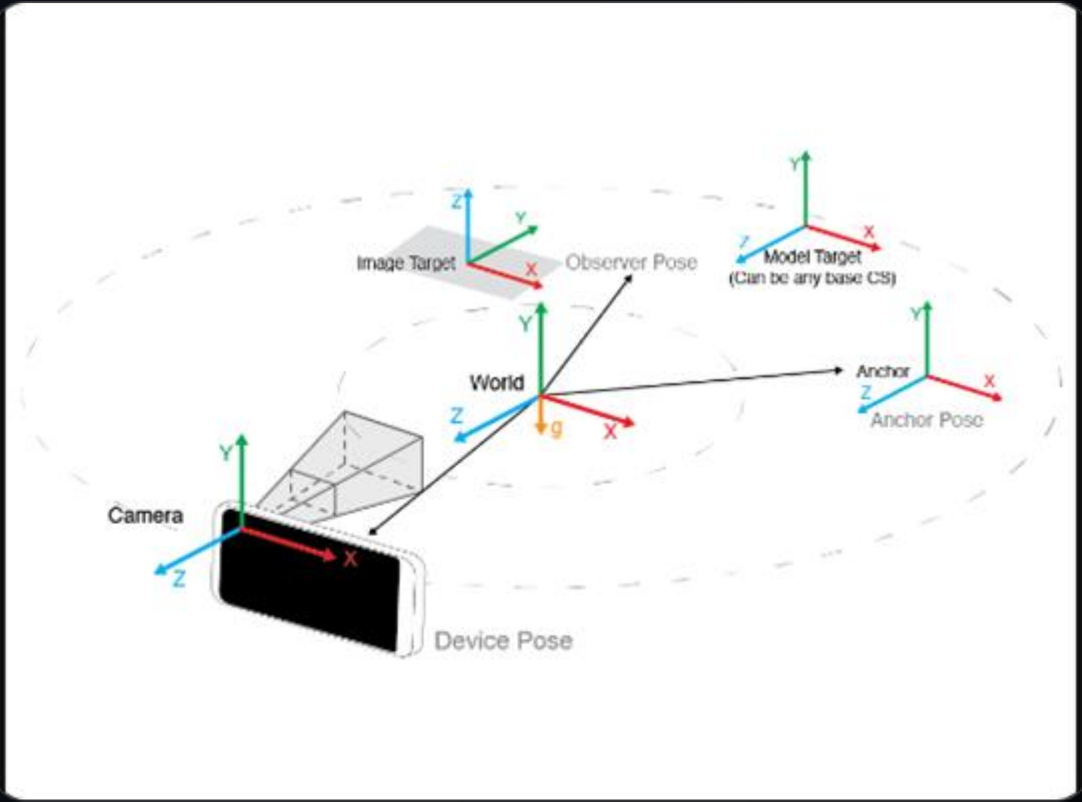
- **Latency:** Delay between motion and display update causing "swimming".
- **Accuracy:** Small errors in pose estimation lead to noticeable misalignment.
- **Robustness:** Handling dynamic environments, occlusion, and lighting changes.
- **Drift:** Accumulation of errors over time in tracking systems.

Coordinate Systems

The Transformations

Registration involves finding the transformation matrices between these spaces:

- **World (C_w):** Fixed global origin.
- **Camera (C_c):** Moving frame attached to the device.
- **Image (C_i):** 2D pixel coordinates.
- **Object (C_o):** Local frame of a virtual object.



Camera Calibration

We model the camera using the Pinhole Camera Model. Calibration determines:

- **Intrinsics (K):** Focal length (f_x, f_y) and principal point (c_x, c_y). Maps 3D camera coords to 2D image plane.
- **Extrinsics ($[R|t]$):** Rotation and Translation. Maps World coords to Camera coords. This is what we track in real-time.
- **Distortion:** Radial and tangential lens distortion coefficients (k_1, k_2, p_1, p_2).

What is Camera Calibration?

- The process of estimating a camera's internal and external parameters.
- Goal: To build a mathematical model that accurately maps 3D world points to 2D image pixels.
- Corrects for lens distortion and determines focal length and principal point.
- Typically involves capturing images of a known calibration pattern (e.g., a checkerboard).
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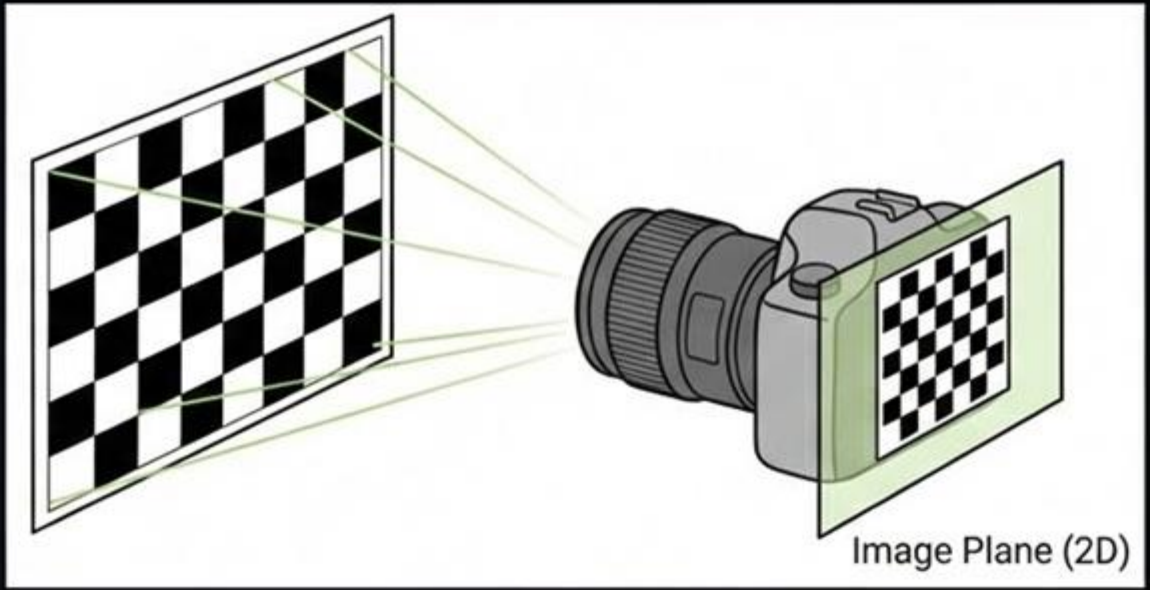
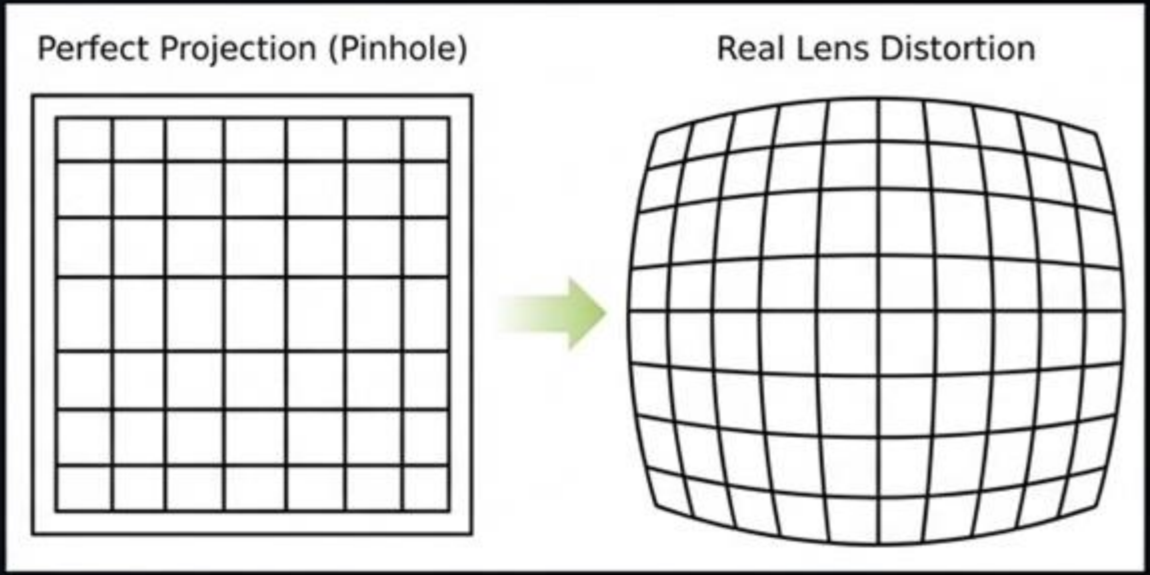


The Challenge: Lens Distortion

- Real cameras are not perfect 'pinhole' devices.
- Lenses bend light, causing geometric distortions in the image.
- Common types: Radial (barrel/pincushion) and Tangential distortion.
- Result: Straight lines in the world appear curved in the image.



The AR Tracking Problem

- Continuous process of estimating the camera's pose (position & orientation) in real-time.
- Must be accurate, robust, and low-latency to maintain registration.
- Challenges include: rapid motion, occlusion, poor lighting, and lack of texture.
- Accumulation of small errors leads to "drift" over time.

Feature Detection

Keypoints

Points of interest in an image that are invariant to scale, rotation, and illumination. Corners and blobs are common features.

Descriptors

A numerical vector describing the local pixel patch around a keypoint. Used to match keypoints across frames.

Visual Odometry (VO)

2D-2D

Motion estimation between two consecutive image frames using epipolar geometry.

3D-2D

Perspective-n-Point (PnP). estimating pose from known 3D points and 2D projections.

3D-3D

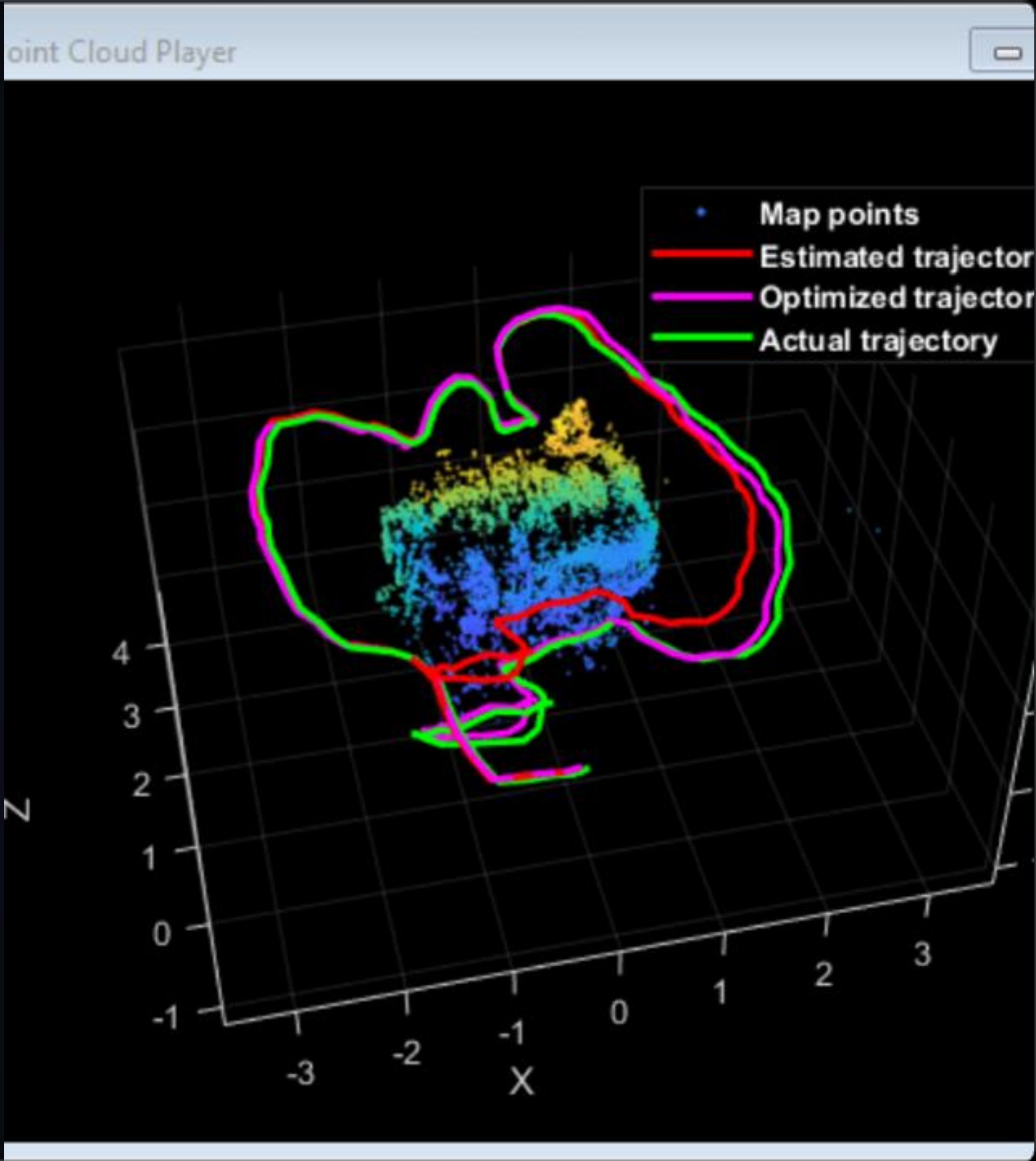
Aligning two 3D point sets

SLAM

Simultaneous Localization and Mapping

VO drifts over time. SLAM adds a "Mapping" thread to reduce drift.

It builds a global map of the environment (sparse feature points or dense mesh) and localizes the camera within it simultaneously.



Tracking Paradigms

Marker-Based

Relies on known fiducial markers (QR codes, ArUco). Robust, fast, and provides absolute scale.

Use case: Industrial maintenance, simple board games.

Markerless (NFT)

Natural Feature Tracking. Uses environmental features (edges, textures). Requires initialization and is computationally heavier.

Use case: Consumer AR, navigation, gaming.

Deep Learning in Tracking

Learned Descriptors

SuperPoint, SuperGlue. replacing hand-crafted descriptors (ORB) with Neural Networks for better matching in challenging lighting.

End-to-End VO

Directly estimating pose from video sequences using RNNs or CNNs, bypassing feature extraction.

PART III

3D Reconstruction

Point Clouds to Meshes

Sparse to Dense

SLAM produces a sparse point cloud. For interactions (occlusion, physics), we need a dense surface.

Meshing Algorithms

Techniques like Delaunay Triangulation or Poisson Surface Reconstruction convert points into a triangular mesh in real-time.

Scene Understanding

- **Plane Detection:** RANSAC (Random Sample Consensus) is used to fit planes to point clouds. Critical for placing objects on floors or walls.
- **Mesh Anchoring:** Attaching virtual content to specific mesh IDs so they persist when the user looks away and returns.
- **Raycasting against Mesh:** Allows virtual objects to "sit" on real surfaces rather than floating.

Semantic Segmentation

Geo

Geometric Understanding: "There is a surface here."

+

Integration

Sem

Semantic Understanding: "This surface is a sofa."

AI models classify pixels or mesh segments, enabling context-aware AR (e.g., virtual cat sits on sofa, not on stove).

PART IV

Illumination & Rendering

"The goal is to render the virtual object such that it is indistinguishable from the real scene."

Photometric Consistency

Light Estimation

Capturing the Environment

To light virtual objects correctly, we need to know the real-world lighting.

- **Spherical Harmonics:** A low-frequency approximation of environmental lighting (ambient + directional).
- **Light Probes:** Reflective spheres captured by the camera to generate environment maps.

Shadows & Grounding

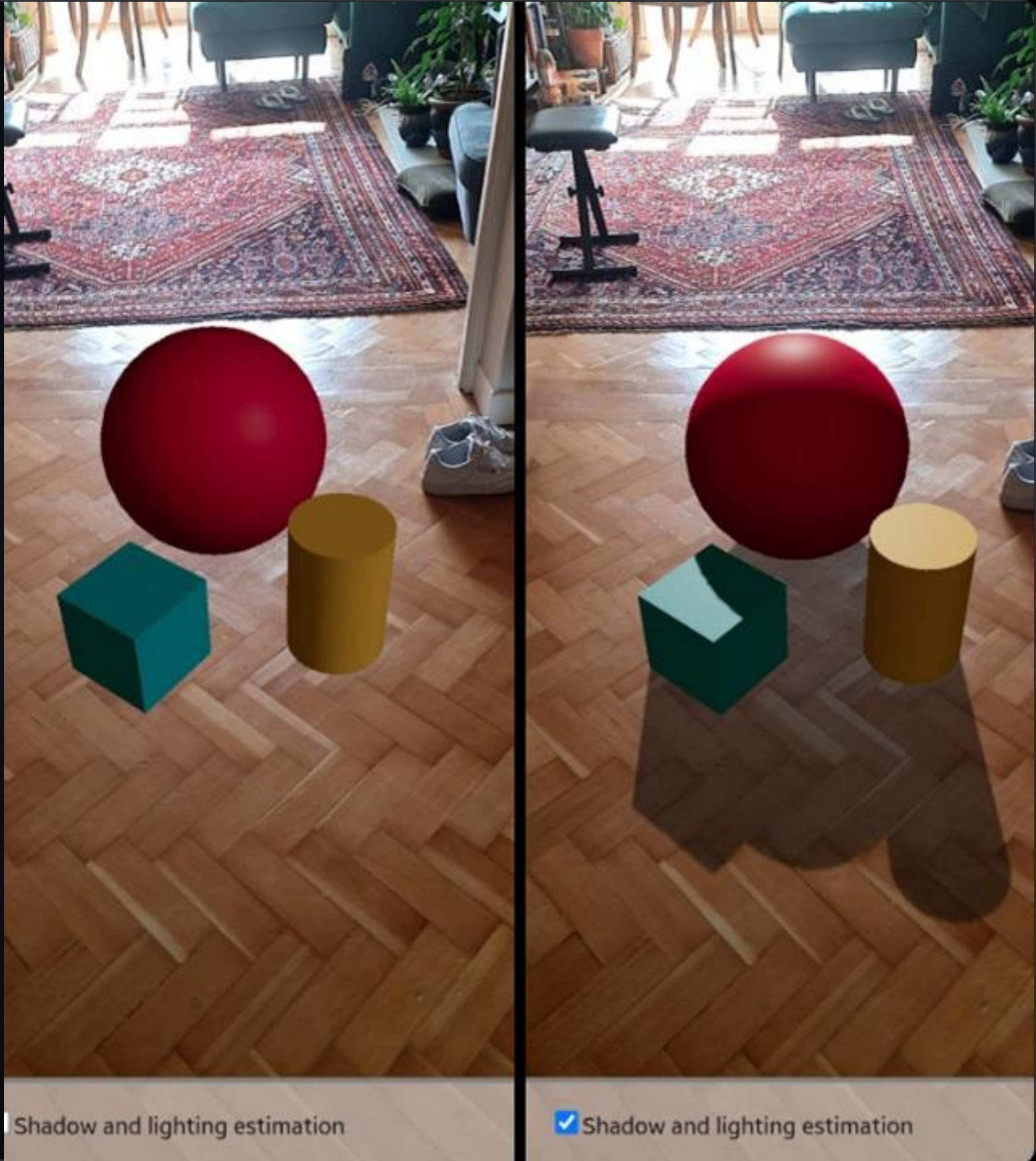
- **Contact Shadows:** Crucial for depth perception. Without shadows, objects appear to float.
- **Shadow Catchers:** Invisible planes placed on real-world surfaces (floors/tables) that only render shadows, not the geometry itself.
- **Soft Shadows:** Ray-traced or blurred shadow maps mimic the area light sources of the real world.

Differential Rendering

Compositing

$$I_{final} = I_{bg} + M \times (I_{virt} - I_{real\_est})$$

Technique to blend virtual and real. We calculate the difference between the rendered scene with virtual objects and the rendered scene of just the background geometry, then apply that delta to the camera feed.



Reflections & Environment Maps

Cube Maps

The camera feed is projected onto a skybox or reflection probe. Glossy virtual objects reflect this texture.

Screen Space Reflections

For high-end AR, we can reflect parts of the real world visible in the camera frame onto virtual surfaces.

Occlusion Handling

Z

Depth Testing

Mask

Depth Masking

Real objects must obscure virtual ones. We render the real-world mesh into the depth buffer but disable color writing. This "phantom geometry" prevents virtual pixels from drawing behind real objects.

PART V

Interaction

Input Modalities

Gesture

Hand tracking (skeleton). Pinch, grab, swipe.

Gaze

Eye tracking. Selection by dwelling or combining with a "click".

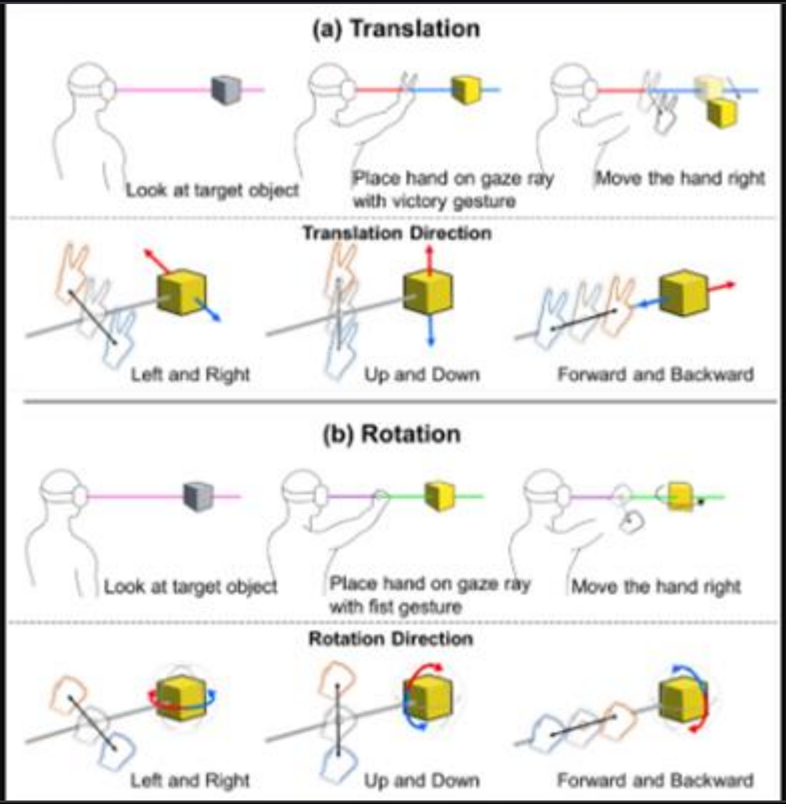
Voice

NLP commands. "Put the cube there."

Interaction Techniques

Selection & Manipulation

- **Ray-Casting:** Projecting a ray from the controller/finger/eye into the scene. First collider hit is selected. Good for distance.
- **Virtual Hand:** Direct manipulation. The user's virtual hand mesh intersects with the object. Good for proximity.



Physics in AR

- **Rigidbodies:** Virtual objects have mass, drag, and gravity.
- **Mesh Colliders:** The real world reconstruction (meshing) is assigned a static mesh collider.
- **Interaction:** When a virtual ball drops, it hits the real floor collider and bounces. This physical consistency is key to immersion.

UI/UX Challenges

World-Fixed vs Screen-Fixed

Should UI float in front of your face (HUD) or be attached to objects (Labels)? HUDs can cause nausea and occlusion issues.

Vergence-Accommodation Conflict

Eyes focus (accommodate) on the screen distance, but converge on the 3D object distance. Mismatch causes fatigue.

PART VI

The Future & XR

Next-Gen Hardware

Form Factor Evolution

Moving from bulky headsets to lightweight glasses and eventually contact lenses.

MicroLED: Extremely bright, self-emissive displays suitable for daylight AR.

Pancake Lenses: Folding light paths to reduce optical depth.

The Metaverse & AI

Social AR

Shared persistent experiences. "Cloud Anchors" allow multiple users to see the same virtual content in the same physical spot.

AI Agents

Context-aware virtual assistants that see what you see and offer help (e.g., identifying parts during engine repair).

Ethical & Privacy Considerations

- **Privacy:** Always-on cameras in public spaces. Facial recognition risks.
- **Information Overload:** Cognitive load management. Danger of distraction (e.g., walking into traffic).
- **Reality Modification:** The power to erase or alter real-world elements (e.g., ad blockers for billboards, or erasing people).

Q&A

Thank you.